

RESEARCH ARTICLE



Analysis of HIV-1 subtypes and drug resistance mutations in people who inject drugs in Aizawl and nearby districts of Mizoram, India

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ABSTRACT

Over the last decade, there has been a steady increase in HIV-1 prevalence in Mizoram, India. Importantly, this increase in HIV-1 prevalence is not only limited to the key population groups such as female sex workers (FSWs) or people who inject drugs (PWID), and has been witnessed in general population as well. Injecting drug use has long been one of the key drivers of HIV-1 epidemic across the north-eastern states of India. In this study, using HIV-1 pol gene region sequences from Aizawl and adjoining districts, we examined the HIV-1 subtypes, recombinant forms, drug resistance mutations and also the spatiotemporal dynamics of the potential unique recombinant forms. In our dataset, the dominant subtype was HIV-1 subtype C (94.91%). We could also identify the presence of CRF01_AE (1.69%) and BC recombinant forms (3.39%). Drug resistance mutation analysis revealed that resistance against non-nucleoside reverse transcriptase inhibitors was most common in the sequences having any resistance mutations. Evolutionary analysis of unique BC recombinants estimated the most recent common ancestor of these sequences around 2004–2005 and them having an ancestry of United States of America (USA) origin.

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Introduction

Since 2000, HIV prevalence in India in adult population (15–49 y) has declined. The estimated HIV prevalence in them in 2000 was at 0.55%, 0.32% in 2010 and 0.21% in 2021. Although the epidemic continues to be of diverse intensity in various states, the north-eastern state of Mizoram has recorded the highest adult HIV prevalence (2.7%), followed by that in Nagaland (1.36%) and Manipur (1.05%) [1]. Under the national free antiretroviral treatment (ART) program, Mizoram provides treatment through 5 ART centres across the state. It is important to note that poor treatment adherence due to different socio-environmental and behavioural factors have been reported resulting in development of HIV-drug resistance.

The HIV Sentinel Surveillance Report-2017 documented highest (19.8%) HIV prevalence among people who inject drugs (PWID) from Mizoram, which was highest among the ones recorded from different states in India [2]. The youths in Mizoram are prone to drug use because of its porous international border with Myanmar and proximity to the Golden-Triangle

(Myanmar, Thailand & Laos) – the site for huge production of illegal production in South-east Asia and the high HIV prevalence among PWID in the state has been attributed to many factors including lack of awareness about safe injection practices, sharing of needles and syringes during drug use, sex work and poverty. The use of shared needles or syringes has been accounted for around 39% of the HIV infections in Mizoram [3,4].

Studies on HIV-1 molecular epidemiology from Manipur, the neighbouring state of Mizoram, revealed subtype C being the dominant form along with the presence of unique recombinant forms. For example, while an earlier study by Bhanja et al. reported the presence of HIV-1 BC recombinant in PWID from Manipur, a relatively recent one by Sharma et al. documented the presence of CRF01_AE, subtype and URF_BC from the wives of PWID in Manipur [5–7]. However, no detailed analysis with respect to HIV-1 genetic diversity from Mizoram is available as yet. Only one study by Clipman et al., among PWID and men having sex with men (MSM) across 21 cities in India,

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addressed the brief phylogenetic clustering and drug resistance profiles for 62 HIV-1 sequences from Aizawl district of Mizoram. However, this study by Clipman et al. discussed the sequences from Aizawl in the context of sequences from the other cities of India [8].

In the present investigation, we undertook phylogenetic, drug resistance mutation and evolutionary analysis of HIV-1 among PWIDs mainly from Aizawl and neighbouring districts of Mizoram. HIV-1 pol gene diversity reflected HIV-1 subtype C being the dominant subtype and the presence of potential unique recombinant forms. Evolutionary estimates indicated that these recombinants might have emerged during 2004–2005, with ancestry of their subtype B traced to be of United States of America (USA) origin. The drug resistance mutation analysis highlighted the presence of resistance mutations in drug naïve individuals indicating the transmitted drug resistance (TDR).

Methods

Study population

This study was carried out in collaboration with Mizoram State AIDS Control Society (MSACS). Dried Blood Spot (DBS) specimens from HIV-1 infected individuals were collected from ART centres and Targeted Intervention sites primarily in Aizawl and surrounding districts. The study was approved by the Ethics Committee, ICMR – National Institute of Translational Virology and AIDS Research (ICMR-NITVAR), Pune, India (Ref: NARI/EC/Approval/20–21/417). Informed written consent was obtained from each of the HIV-1 infected individuals for their participation in this investigation. Collected DBS specimens were shipped from Mizoram to ICMR-NARI, Pune in the western state of Maharashtra for subsequent genotyping.

Amplification and sequencing of HIV-1 pol gene region

DBS specimens were subjected to Total Nucleic Acid extraction (TNA) using NucliSens EasyMag total nucleic acid extraction system (Biomerieux). HIV-1 Pol region encoding Protease gene codons (1–99, PR) and partial Reverse Transcriptase gene codons (1–345, RT) were amplified using in-house genotyping protocol using primers (Supplementary Table S3) mentioned in the WHO manual for HIV drug resistance testing with dried blood spot specimens [9]. Using this strategy, we could successfully amplify the protease and partial

reverse transcriptase region for 59 specimens. For integrase gene amplification (HXB2 coordinates 4230–5092), the same RNA was subjected to in-house amplification and genotyping protocol using primers mentioned in Supplementary Table S3. Out of 59 specimens for which protease-reverse transcriptase region was sequenced, only 49 could be successfully amplified and sequenced for integrase coding region. The amplified products were then purified using QIAquick PCR purification kit (Qiagen). Further, these purified products were utilized for cycle sequencing PCR using BigDye Terminator v3.1 Cycle sequencing kit (Thermo Fisher Scientific). Sequencing was performed with the help of 3130XL Genetic Analyzer (Applied Biosystems). SeqScape software v2.7 was used for sequence contig generation, base-calling and final FASTA file generation.

HIV-1 subtyping, phylogenetic analysis

For subtyping, we used the protease-reverse transcriptase sequences ($n = 59$). These sequences spanning HXB2 coordinates 2253–3454 were subjected to HIV-1 subtype analysis using REGA HIV-1 Subtyping Tool version 3.0 (<http://dbpartners.stanford.edu:8080/RegaSubtyping/stanford-hiv/typingtool/>) [10], COMET HIV-1 (Context-based Modelling for Expeditious Typing) (<https://comet.lih.lu/>) [11], Recombinant Identification Program (RIP) (<https://www.hiv.lanl.gov/content/sequence/RIP/RIP.html>) [12] and jpHMM-HIV-1 (http://jphmm.gobics.de/submission_hiv) [13], respectively. Following this strategy, an individual sequence was assigned a particular subtype where at least 3 out of 4 employed methods showed a concordance. In cases, where any of the platform was unable to assign the subtype, the call was based on the remaining platforms provided they showed a concordance.

For phylogenetic analysis, using protease-reverse transcriptase sequences as query details of the sequences closely related to these sequences were retrieved with the help of ViroBLAST tool (<https://indra.mullins.microbiol.washington.edu/viroblast/docs/aboutviroblast.html>) [14]. The accession numbers of the closely related sequences were used to search specific sequences using Los Alamos HIV-1 sequence database; only one sequence per patient was retained after the search results. These reference sequences ($n = 2272$) along with the study sequences ($n = 59$) spanning HXB2 coordinates 2253–3454 were used as the final dataset for maximum-likelihood phylogenetic tree construction. We also included HXB2 sequence (sequence coordinates: 2253–3454) as an out-group during analysis. This final dataset constituting 2332 sequences was subjected to the removal of major drug resistance coding sites. The phylogenetic tree was constructed using general time

reversible plus gamma (GTR+G) nucleotide substitution model and 1000 bootstrap replicates with the help of standalone IQ-Tree version 2.1 [15]. The final tree was visualised using FigTree version 1.4.4.

Recombinant sequence analysis

For recombinant sequences recombination breakpoints were derived using jpHMM HIV-1 tool. Recombination breakpoint details were further used for depiction of recombinants using the Recombinant HIV-1 Drawing Tool (https://www.hiv.lanl.gov/content/sequence/DRAW_CRF/recom_mapper.html). Further, the recombination patterns of the study sequences were compared with the already designated CRF's.

Drug resistance mutation interpretation

For drug resistance mutation (DRM) analysis out of 59 sequences, we only considered 49 sequences as for these sequences both regions covering protease-reverse transcriptase (HXB2 coordinates 2253–3454) and integrase (HXB2 coordinates 4230–5092) were sequenced. These sequences ($n = 49$) were analysed with the help of HIVdb Program, Genotypic Resistance Interpretation Algorithm available at the Stanford University HIV Drug Resistance Database (<https://hivdb.stanford.edu/hivdb/by-sequences/>).

Estimation of HIV-1 BC recombinant origin location and time

We employed the local instance of Nextstrain to analyse the geographical ancestry of the recombinant sequences. Nextstrain implements the Augur bioinformatics pipeline to subsample data, align sequences, phylogeny reconstruction followed by estimation of phylogeographic patterns which could be visualized with Auspice (<https://auspice.us/>) [16,17]. The analysis was restricted to the subtype B portion of the recombinant sequences spanning HXB2 coordinates 2253–2835 since this was common to both the BC recombinants. The Nextstrain HIV-1 subtype B sequence alignment metadata corresponding to the pol gene region sequences was downloaded from Nextstrain portal (<https://nextstrain.org/groups/LANL-HIV-DB/HIV/pol>). The GenBank accession numbers available in the metadata were used to download these sequences (HXB2 coordinates: 2253–2835, $n = 1124$) from HIV sequence database (hiv.lanl.gov). Further, these sequences were combined with the HIV-1 recombinant sequences (22–9575 & 22–9659) from this study. We also included the relevant metadata of two study sequences in the Nextstrain metadata file. The final sequence alignment

($n = 1126$) and corresponding metadata were then used for ancestral state reconstruction using Nextstrain workflow.

Bayesian discrete trait analysis

In addition to the Nextstrain analyses, we also inferred upon the ancestry of the HIV-1 BC recombinants using discrete Bayesian phylogeographic method accompanied with Bayesian Stochastic Search Variable Selection (BSSVS) in BEAST v 1.10.4. To reduce the computational load, for this analysis we only used the sequences within the clade containing study recombinant sequences from an initial Newick tree generated during the Nextstrain workflow. This dataset constituted the 49 sequences representing Belgium ($n = 2$), Brazil ($n = 4$), China ($n = 2$), Cyprus ($n = 1$), Denmark ($n = 1$), Germany ($n = 4$), Great Britain ($n = 6$), Japan ($n = 3$), Peru ($n = 1$), Russia ($n = 1$), South-Korea ($n = 1$), Sweden ($n = 1$), Switzerland ($n = 1$), United States of America ($n = 19$) and two study sequences from India. Country locations of the respective sequences were assigned as the discrete traits. Bayesian maximum clade credibility tree with discrete traits was constructed using GTR+G+I substitution model, uncorrelated relaxed lognormal molecular clock and skyline tree prior. The BEAST analysis was performed for 20 million steps with sampling at every 2000 steps along with the implementation of BEAGLE library [18]. The annotated MCC tree was generated using the Tree Annotator version 1.10.4 with the burn-in of 10%. The final MCC tree with discrete traits and log file from BSSVS analysis were used to summarize the viral migration routes using “Spatial Phylogenetic Reconstruction of Evolutionary Dynamics” software using Data-Driven Documents (D3) (Spread3 v0.97) [19]. The country rates log file was used to calculate the Bayes factor (BF) support for the viral spread between countries and diffusion rates having $BF > 3$ were considered as well-supported diffusion rates [20].

Sequence data

Nucleotide sequences were submitted to GenBank under accession numbers for protease & reverse transcriptase (PR-RT) region: OR834866–OR834924. For integrase (IN) region the GenBank accession numbers are OR834925–OR834983.

Results

Demographic characteristics of participants

As shown in Table 1, all the sequences generated in this study belonged to the individuals who inject drugs. Majority of the individuals ($n = 47$) were the residents of Aizawl district, followed by 8 from Lawngtlai district

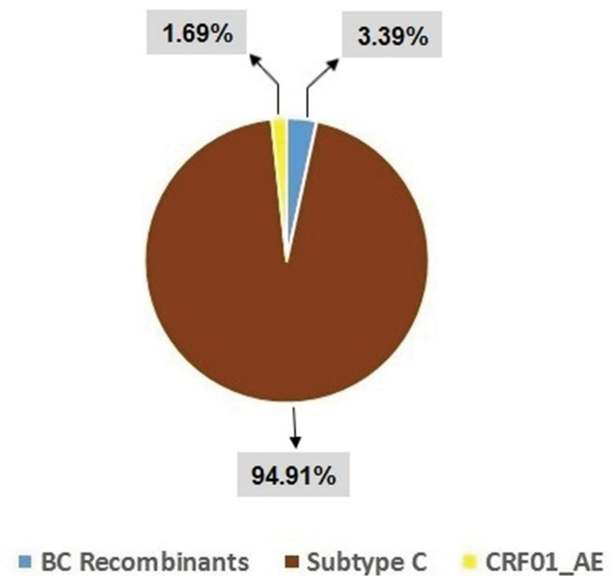
Table 1. Demographic characteristics of study participants.

Characteristic	Sample Size (N = 59)
Gender	
Male	55
Female	3
Unknown	1
Median age (year)	30 (19–47 y)
Districts	
Aizawl	47
Lawngtlai	8
Siaha	2
Lunglei	2
Treatment history	
Naïve	40
Treated	19

and 2 each from Lunglei and Siaha districts. Out of 59 cases, 55 were male, 3 were female and for 1 case the gender data was not available. The median age of these participants was 30 y (for one case no age data was available). Out of the 59 individuals, 40 were ART-drug naïve, while the remaining 19 were on antiretroviral treatment. All 19 individuals on ART had received a dolutegravir-based regimen comprising tenofovir, lamivudine and dolutegravir (TLD) for 12 months as on the date of specimen collection.

HIV-1 *pol* gene diversity

A total of 59 sequences of PR-RT region, which covered HXB2 coordinates 2253–3454, were subjected to subtype analysis (Supplementary Table S1). Out of 59 sequences, 56 (94.91%) were observed to be of HIV-1 subtype C (Figure 1). Out of the remaining three sequences, 1 (1.69%) was CRF01_AE, and remaining 2 (3.39%) were not assigned any pure subtype or circulating recombinant form (CRF) nomenclature; instead, they were categorized as the BC recombinants. Maximum-likelihood phylogenetic tree (Supplementary Figure S1) constructed as described in the methods revealed that the majority of the HIV-1 C study sequences were part of the HIV-1 C clade containing sequences representing India and China. One of the study sequence, which was subtyped as CRF01_AE, was observed to be grouping with CRF01_AE sequences primarily from Thailand and China within the clade consisting CRF01_AE sequences. Two sequences, which were not assigned any specific known subtype (22–9575 and 22–9659), were placed near the root of the tree close to HXB2 sequence and BC recombinant sequences from China (Supplementary Figure S1).

**Figure 1.** Pie chart representing the proportion (%) of HIV-1 subtypes based on the *pol* gene sequences ($n = 59$).

Drug resistance mutations

Out of 59 sequences available for protease-reverse transcriptase region only 49 could be genotyped for integrase region. Further, these 49 sequences (Supplementary Table S2) for which both the regions were sequenced were subjected to drug resistance mutation interpretation. Out of these 49 individuals 37 were ART-drug naïve while remaining 12 were receiving the TLD regimen. As shown (Table 2), out of 37 drug naïve individuals 12 (32.43%) had the drug resistance mutations either against protease inhibitor (PI) or nucleoside reverse transcriptase (NRTI) and/or non-nucleoside reverse transcriptase inhibitors (NNRTI). Out of these 12 sequences majority had the DRMs against NNRTIs ($n = 9$). K103N ($n = 3$), E138AK ($n = 3$), V179DE ($n = 3$) were the most common DRMs against NNRTIs, followed by the presence of Y181C and H221Y present individually in two sequences. Only 3 sequences were observed to have the presence of DRMs (I54T, I84V and L90M) against PI and two were having the DRMs A62V and S68N against the NRTIs. None of the drug naïve sequences had the DRMs against integrase inhibitor.

In case of drug treated individuals ($n = 12$) five (41.67%) had any DRM and all of these had the

Table 2. Drug resistance mutations.

	Sequences N	Any DRM N (%)	PI DRM N (%)	NRTI DRM N (%)	NNRTI DRM N (%)	IN DRM N (%)
Cumulative	49	17 (34.69)	4 (8.16)	2 (4.08)	14 (28.57)	0 (0.00)
Drug Naïve	37	12 (32.43)	3 (8.10)	1 (2.70)	9 (24.32)	0 (0.00)
Drug Treated	12	5 (41.67)	1 (8.33)	1 (8.33)	5 (41.67)	0 (0.00)

In Table 2 DRM data represents only 49 out of 59 sequences for which PR-RT as well as IN gene regions were amplified.

DRMs against NNRTIs. In case of these five sequences, the most frequent NNRTI DRM was V179D present in 3 sequences followed by E138AK observed in 2 sequences. Only two out of these five sequences had the DRM against PI (M46L) or NRTI (S68N). None of the 12 treatment experienced individuals showed the presence DRMs against integrase inhibitor.

HIV-1 BC recombinants breakpoint analysis

As illustrated in Figure 2a, recombination pattern analysis of the two BC recombinants revealed that for sequence 22-9575 the pol gene fragment spanning HXB2 coordinates 2253–3178 represented the HIV-1 subtype B, while the region between HXB2 coordinates 3179–3454 represented the subtype C. In case of sequence 22-9659, the

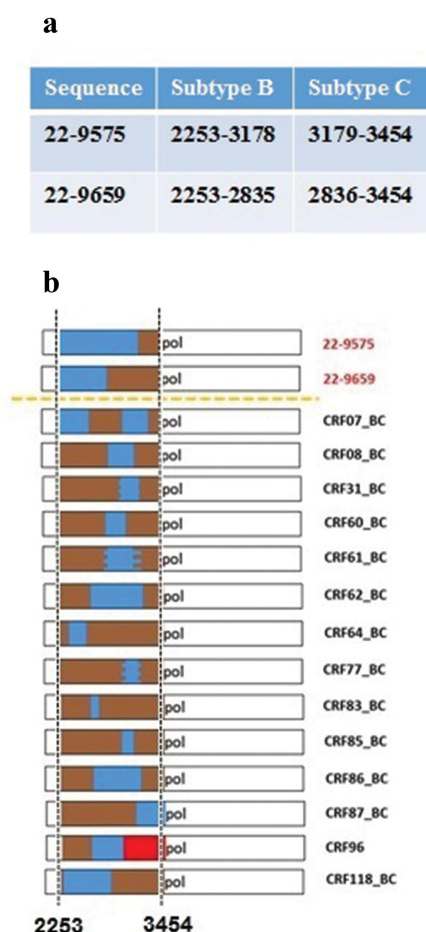


Figure 2. (a) Recombination breakpoints for the BC recombinants with respect to the HXB2 coordinates. (b) Cartoon representing the recombination breakpoints for sequences from Mizoram (22-9575 & 22-9659) against the CRF's having evidence of subtype B and C recombination between the HXB2 coordinates 2253–3454.

region spanning HXB2 coordinates 2253–2835 was HIV-1 subtype B, while the portion beyond HXB2 coordinates 2836 till 3454 was subtype C. On comparing these two sequences against the known CRFs (Figure 2b) having HIV-1 B and C recombination within the HXB2 coordinates 2253–3454, it was observed that the recombination pattern of these sequences was different compared to those documented CRFs indicating these being potential unique recombinant forms (URF).

Spatiotemporal analysis of BC recombinant ancestry

Evolutionary analysis of these potential unique BC recombinants using Nextstrain (Figure 3) identified that the subtype B fragment of both the recombinants shared the same node and the time to the most recent common ancestor (TMRCA) for these sequences was observed to be July 2004 (Date Confidence Interval: February 1999–January 2009). Nextstrain analysis highlighted USA as the most probable geographical source for these sequences.

Further to elaborate upon these findings, we carried out the Bayesian discrete trait reconstruction using BSSVS analysis in BEAST 1.10.4 as described in methods. The dataset for Bayesian analysis was compiled as described in the methods. The regression analysis carried out using TempEst to estimate the correlation between tip dates and root to tip genetic divergence revealed strong clock-like pattern for the used dataset. According to TempEst, the substitution rate estimated for this dataset was 1.075×10^{-3} substitutions per site per year with the correlation coefficient of 0.52 and r^2 of 0.27. In Bayesian MCC tree (Figure 4a) generated, the subtype B fragment of BC recombinants shared the same node and the TMRCA calculated for this node was observed to be 2005.12 (95% HPD: 1993.60–2014.95) with mean substitution rate of 1.075×10^{-3} (95%HPD: 6.012×10^{-4} – 1.54×10^{-3}). The viral migration events reconstructed using Spred3 (Figure 4b) identified USA as the origin for these sequences with the strong Bayes factor support (BF = 24.81) for diffusion from USA to India.

Discussion

According to the 2020-estimates, adult HIV prevalence among adults (15–49 y) in the state of Mizoram was 2.37%, while that in the Aizawl district was 3.99%, highlighting an epicentre phenomenon [21]. HIV prevalence in PWID (19.8%) in the state has been reported to be considerably higher than the national average [4,22] recorded in the same key population group.

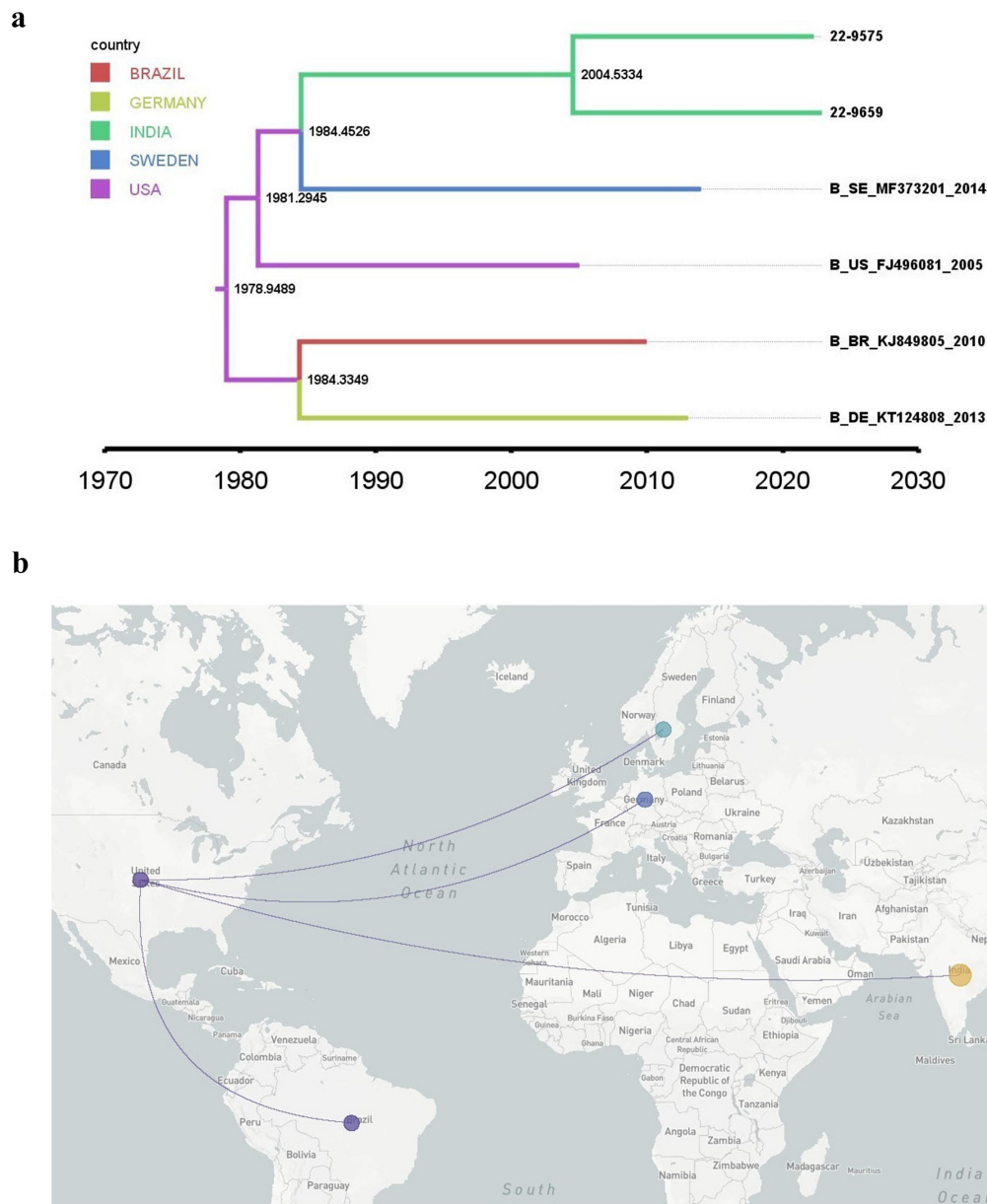


Figure 3. (a) a node from a phylogenetic tree constructed with the help of nextstrain platform displaying the clustering and TMRCA of subtype B fragments for BC recombinants. (b) A geographical representation of spatial dynamics for BC recombinants visualized with the help of auspice.

Despite such high HIV prevalence in the state and its link with PWID, little was explored till date pertaining to the molecular epidemiology of HIV among PWID.

Molecular phylogenetic analysis of HIV-1 pol gene region from PWIDs from the Aizawl, Lawngtlai, Lunglei and Siahla districts during present investigation highlighted HIV-1 C as the dominant subtype. The porous nature of international border shared by Mizoram with Myanmar has been documented to be a major gateway (3) for illegal drug trade and human migration. Phylogenetic analysis revealed the presence

of potential unique HIV-1 BC recombinants clustering close to the unassigned BC recombinant sequences from China and also the CRF01_AE sequences closely resembling the CRF01_AE from China. The phylogeographic estimates derived using Nextstrain and Bayesian reconstruction of subtype B fragments for BC recombinants revealed the shared ancestry for both the recombinants and it was estimated to be of USA origin. These observations reflect upon a possibility of introduction of USA origin HIV-1 subtype B in this part of the country followed by the

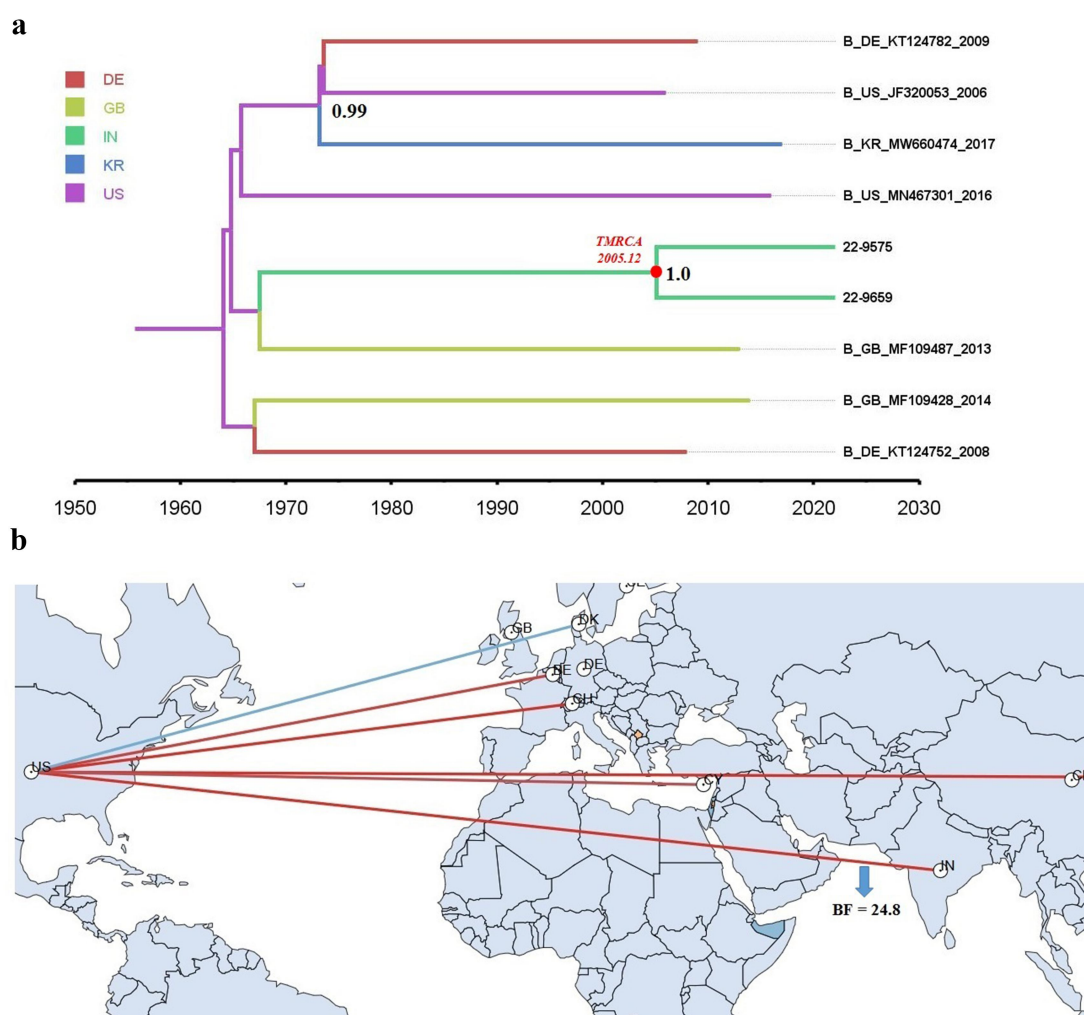


Figure 4. (a) Node form a bayesian MCC tree showing the TMRCA for BC recombinant subtype B fragments. (b) Geographical reconstruction of the discrete trait analysis performed in Spred3. Blue arrow indicates the Bayes factor support for viral diffusion from USA to India.

recombination events with the dominant subtype C. The evolutionary estimates for these sequences using Nextstrain and Bayesian analysis were also similar, which indicated that these recombinants might have emerged around 2004–2005.

Our analysis highlighted the presence of transmitted drug resistance mutations in drug naïve individuals. K103N is known to be most commonly transmitted drug resistance mutation and its presence prior to starting first-line regimens containing NNRTIs is associated with elevated chances of virologic failure [23]. In our analysis also K103N was one of the common TDR mutation indicating the poor outcome in case these individuals to be treated with NNRTI containing regimens. Mizoram where injecting drug use has been documented to have an interface with unsafe sexual practices [3,4], may contribute to the rapid spread of

resistant viral strains to the other high risk and general population. The presence of such TDR strains within PWID networks warrants immediate detailed investigations while strengthening the ongoing HIV-treatment and prevention programs and practices. However, there were no DRMs against integrase inhibitors either in naïve or treated individuals.

In recent times, Mizoram has experienced a surge in HIV prevalence among PWID, FSWs and general population. Nearly, 39% of the newly diagnosed HIV cases in Mizoram had been attributed to needle sharing practices during drug use. Additionally, the unsafe sexual practices contributed to the spread of HIV to general population [3,4]. Persistent molecular surveillance aided with cluster analysis would enable tracking of possible HIV outbreaks in geographical locations and thereby enabling timely public health action to

intercept occurrence of new infections [24,25]. Ongoing molecular surveillance assisted with cluster analysis could be useful in identifying the localities having high PWID population and preventing the possibility of potential HIV outbreaks, especially detecting the clusters having the potential to spread HIV-drug resistant strains. In combination with contact tracing, sequence-based transmission cluster construction as well as evolutionary analysis can provide pivotal information regarding HIV-transmission dynamics and inform intervention initiatives that could prevent potential outbreaks.

Mizoram, the north-eastern state of India, shares its borders with the neighbouring countries of Myanmar and Bangladesh where the presence of non-subtype C and recombinant HIV-1 forms have been well documented [26–28]. As these borders are porous, it would be interesting to examine HIV-1 strain diversity in the districts of Mizoram adjacent to these borders. Presence of potential URFs in Mizoram based on HIV-1 pol gene region warrants full-length genome sequencing of these viruses to understand their genomic architecture beyond pol gene.

In summary, our study could provide the snapshot of HIV-1 molecular epidemiology in PWIDs from Aizawl and to some extent surrounding districts. The main limitation of our study is a small sample size, nonetheless current investigation provides the necessary primer for future directions to elaborate on these findings. With better surveillance strategies comprising appropriate sampling frequencies across the multiple districts would certainly aid in exploring further details, especially among key populations like PWIDs in Mizoram, where injecting drug uses is one of the key risks leading to HIV transmissions.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The DNA nucleotide sequence data that support the findings of this study are openly available in GenBank at <https://www.ncbi.nlm.nih.gov/genbank> with accession numbers OR834866–OR834924 for protease & reverse transcriptase (PR-RT) region and OR834925–OR834983 for integrase (IN) region.

Ethical statement

The study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki. The study received an approval from the Ethics Committee at ICMR-National Institute of Translational Virology and AIDS Research (ICMR-NITVAR) in Pune, India (Ref: NARI/EC/Approval/20–21/417). Written informed consent was acquired from each participant before their involvement in the research.

Author contributions

KR: Data collection, analysis, data interpretation, and manuscript writing. LT: Analysis, review, and supervision. AL: data collection and analysis. DK: Data collection, analysis, and interpretation. AP: Phylogenetic analysis, spatiotemporal analysis using Nextstrain bioinformatics pipeline, Bayesian discrete trait analysis using the BEAST platform, data interpretation, and manuscript writing. PC: Analysis, data interpretation, and manuscript writing. RCH: Data collection and analysis. SP: Study planning, review, editing, and supervision. SK: Conceived the study, funding acquisition, analysis, review and supervision. All authors have contributed to the manuscript and approved the submitted version.

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